

Balraj Ashwath

AI Researcher / Data Scientist | [in](#) LinkedIn | [g](#) Google Scholar | [k](#) Kaggle

✉ balrajashwath98@gmail.com

☎ +91-9964952869

EDUCATION

- **National Institute of Technology Karnataka** Surathkal, IN
B.Tech in Electronics and Communications Engineering; GPA: 7/10 | Senior Thesis: 10/10 Aug. 2015 – Apr. 2019
- **Narayana Pre-University College** Bengaluru, IN
Karnataka State Pre-University; Percentage: 90% | JEE Mains: 176/360 (99th Percentile) Jul. 2013 – May. 2015

EXPERIENCE

- **Microland Limited** Bengaluru, IN
Data Scientist / ML Engineer Jul. 2019 - Present
 - **Text-based Analysis & Modeling:** Worked on natural language processing pipelines for tasks such as unsupervised text clustering, topic modeling, text classification & sentiment analysis.
 - **Server Metric Analysis & Modeling:** Worked on time series forecasting & anomaly detection on server metrics. Developed end-to-end ML pipelines with microservices-based architecture.
 - **Tech Stack:** TensorFlow, PyTorch, Gensim, Elasticsearch, Kibana, Scikit-learn, Gensim, SpaCy, Pandas, Prophet
- **Video Analytics Lab, CDS Department, Indian Institute of Science** Bengaluru, IN
Research Intern (Mentors: Dr. R. Venkatesh Babu & K. Ram Prabhakar) May. 2019 – Jul. 2019
 - **High-resolution HDR deghosting using CNNs:** Worked on High Dynamic Range motion detection & correction. Developed & implemented novel deep-learning method to generate high-quality fused HDR image that achieved SOTA performance in terms of accuracy & speed [Work accepted for poster presentation at ECCV 2020]
- **Image Processing & Analysis Lab, National Institute of Technology Karnataka** Surathkal, IN
Undergraduate Researcher (Mentors: Dr. Shyam Lal & Karuna K. Eerapu) Jul. 2018 – Apr. 2019
 - **Deep Learning based approaches for Road Extraction from Satellite Imagery:** Developed novel semantic segmentation architecture to efficiently share higher-level image features across various resolution scales. Our model achieved SOTA performance in road accuracy on Massachusetts Roads dataset [Journal accepted at IEEE Access]
- **RBL Bank** Mumbai, IN
Computer Vision & Machine Learning Intern May 2018 – Jul. 2018
 - **e-KYC process Automation using CNNs for Optical Character Recognition on PAN/ Aadhar cards:** Developed CNN-based OCR models achieving 93% mean accuracy score & sub-40ms inference time per document.

PUBLICATIONS

- **Dense Refinement Residual Network for Road Extraction from Aerial Imagery Data**, Karuna K. Eerapu*, Balraj Ashwath*, Shyam Lal, Fabio D. Acqua & A.V. Narasimhadhan, *IEEE Access Journal*, 2019. [* - equal contbⁿ]
- **Towards Practical and Efficient High-Resolution HDR Deghosting with CNN**, K. Ram Prabhakar, Susmit Agrawal, Durgesh Singh, Balraj Ashwath, & R. Venkatesh Babu, *European Conference on Computer Vision*, 2020.

PROGRAMMING SKILLS

ML Frameworks: PyTorch, TensorFlow, Keras | **Languages:** Python, Java, C, C++, R, Matlab, SQL, $\LaTeX$

ACTIVITIES / INTERESTS / VOLUNTEER EXPERIENCE

- Reading popular research papers & blog posts. Participating in Kaggle competitions & writing Kaggle notebooks
- Math/Science Tutor & Fundraising Volunteer at Prerana (NGO). Coordinated activities at Voice Club, NITK